

Algebra Essentials

ANSWER KEY



Evaluating expressions

- 1 If $x = 4$, evaluate $3x + 7$.

$$3(4) + 7 = 12 + 7 = 19.$$

- 2 If $a = 5$ and $b = 2$, evaluate $a^2 - 3b$.

$$5^2 - 3(2) = 25 - 6 = 19.$$

- 3 If $x = -3$, evaluate $2x^2 + 5x$.

$$2(-3)^2 + 5(-3) = 2(9) - 15 = 18 - 15 = 3.$$

- 4 If $p = 6$ and $q = -2$, evaluate $\frac{p - q}{q + 6}$.

$$\frac{6 - (-2)}{-2 + 6} = \frac{8}{4} = 2.$$

Solving linear equations

- 5 Solve $2x + 5 = 17$.

$$2x = 12, \text{ so } x = 6.$$

- 6 Solve $5x - 8 = 27$.

$$5x = 35, \text{ so } x = 7.$$

- 7 Solve $3(x - 4) = 21$.

$$x - 4 = 7, \text{ so } x = 11.$$

- 8 Solve $\frac{x}{4} + 3 = 10$.

$$\frac{x}{4} = 7, \text{ so } x = 28.$$

- 9 Solve $7x + 3 = 4x + 18$.

$$7x - 4x = 18 - 3, \text{ so } 3x = 15, \text{ and } x = 5.$$

Expanding and simplifying

- 10 Expand $(x + 3)(x + 5)$.

$$x^2 + (3 + 5)x + 3 \times 5 = x^2 + 8x + 15.$$

- 11 Expand $(x - 4)(x + 6)$.

$$x^2 + (-4 + 6)x + (-4)(6) = x^2 + 2x - 24.$$

- 12 Expand $(2x + 1)(x - 3)$.

$$2x^2 - 6x + x - 3 = 2x^2 - 5x - 3.$$

- 13 Simplify $3(2x - 5) - 2(x - 4)$.

$$6x - 15 - 2x + 8 = 4x - 7.$$

Factoring

- 14 Factor $x^2 + 9x + 20$.

Find two numbers with sum 9 and product 20: 4 and 5. $(x + 4)(x + 5)$.

- 15 Factor $x^2 - 2x - 15$.

Find two numbers with sum -2 and product -15 : -5 and 3 . $(x - 5)(x + 3)$.

- 16 Factor $x^2 - 49$.

Difference of squares: $(x - 7)(x + 7)$.

- 17 Factor $6x + 15$.

Common factor 3: $3(2x + 5)$.

Solving quadratic equations by factoring

- 18 Solve $x^2 + 5x + 6 = 0$.

Factor: $(x + 2)(x + 3) = 0$, so $x = -2$ or $x = -3$.

- 19 Solve $x^2 - 7x + 12 = 0$.

Factor: $(x - 3)(x - 4) = 0$, so $x = 3$ or $x = 4$.

- 20 Solve $x^2 - 16 = 0$.

Difference of squares: $(x - 4)(x + 4) = 0$, so $x = 4$ or $x = -4$.

21 Solve $x^2 + 2x - 15 = 0$.

Factor: $(x + 5)(x - 3) = 0$, so $x = -5$ or $x = 3$.

Systems of two equations

22 Solve the system: $x + y = 10$, $x - y = 4$.

Adding the equations: $2x = 14$, so $x = 7$. Then $y = 10 - 7 = 3$.

23 Solve the system: $2x + y = 11$, $x - y = 1$.

Adding the equations: $3x = 12$, so $x = 4$. Then $y = x - 1 = 3$.

24 Solve the system: $x = 3y - 1$, $2x + y = 19$.

Substitute: $2(3y - 1) + y = 19 \Rightarrow 6y - 2 + y = 19 \Rightarrow 7y = 21 \Rightarrow y = 3$. Then $x = 3(3) - 1 = 8$.

25 Solve the system: $4x - y = 10$, $2x + y = 8$.

Adding the equations: $6x = 18$, so $x = 3$. Then $y = 8 - 2(3) = 2$.

26 Solve the system: $5x + 3y = 21$, $2x - 3y = 0$.

Adding the equations: $7x = 21$, so $x = 3$. Then $3y = 2x = 6$, so $y = 2$.

Laws of exponents

27 Simplify $a^5 \times a^3$.

$$a^{5+3} = a^8.$$

28 Simplify $\frac{x^9}{x^4}$.

$$x^{9-4} = x^5.$$

29 Simplify $(y^3)^4$.

$$y^{3 \times 4} = y^{12}.$$

30 Simplify $2x^3 \times 5x^2$.

$$2 \times 5 \times x^{3+2} = 10x^5.$$

Inequalities

31 Solve $3x - 4 > 11$.

$$3x > 15, \text{ so } x > 5.$$

32 Solve $-2x + 6 \leq 20$.

$$-2x \leq 14. \text{ Dividing by a negative reverses the inequality: } x \geq -7.$$

33 Solve $5x + 1 \geq 3x + 9$.

$$2x \geq 8, \text{ so } x \geq 4.$$

34 Solve $4(x - 2) < 2x + 6$.

$$4x - 8 < 2x + 6 \Rightarrow 2x < 14 \Rightarrow x < 7.$$

Substitution into formulas

35 The formula for simple interest is $I = Prt$. Find I when $P = 2,000$, $r = 0.04$, $t = 3$.

$$I = 2,000 \times 0.04 \times 3 = 240.$$

36 The perimeter of a rectangle is $P = 2(l + w)$. Find P when $l = 15$ and $w = 9$.

$$P = 2(15 + 9) = 2(24) = 48.$$

37 If $v = u + at$, find v when $u = 5$, $a = 3$, $t = 4$.

$$v = 5 + 3(4) = 5 + 12 = 17.$$

38 The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 14$ and $h = 9$.

$$A = \frac{1}{2} \times 14 \times 9 = 63.$$

Translating words into equations

39 Twice a number, plus 7, equals 25. Find the number.

$$2x + 7 = 25 \Rightarrow 2x = 18 \Rightarrow x = 9.$$

40 A number decreased by 8 is equal to three times the number, decreased by 20. Find the number.

$$x - 8 = 3x - 20 \Rightarrow 12 = 2x \Rightarrow x = 6.$$

41 The sum of three consecutive integers is 84. Find the integers.

$$\text{Let the integers be } n, n + 1, n + 2: 3n + 3 = 84 \Rightarrow 3n = 81 \Rightarrow n = 27. \text{ The integers are } 27, 28, 29.$$

- 42 Five more than three times a number is the same as 10 less than five times the number. Find the number.

$$3x + 5 = 5x - 10 \Rightarrow 15 = 2x \Rightarrow x = 7.5.$$

Word problems

- 43 A rectangle has length $(x + 3)$ cm and width x cm. This question has two parts.

a If the area is 40 cm^2 , write an equation in x . Area = $x(x + 3) = 40$, i.e. $x^2 + 3x - 40 = 0$.

b Solve for x . Factoring: $(x + 8)(x - 5) = 0$. Since length must be positive, $x = 5$.

- 44 A father is 3 times as old as his son. In 12 years, he will be twice as old as his son. Find their current ages.

Let the son's age be x ; the father's is $3x$. In 12 years:

$$3x + 12 = 2(x + 12) \Rightarrow 3x + 12 = 2x + 24 \Rightarrow x = 12. \text{ Son is 12, father is 36.}$$

- 45 Tickets cost 25 riyals for adults and 15 riyals for children. A group of 10 people paid 220 riyals in total. How many adults were in the group?

Let a be the number of adults, so $10 - a$ are children:

$$25a + 15(10 - a) = 220 \Rightarrow 25a + 150 - 15a = 220 \Rightarrow 10a = 70 \Rightarrow a = 7 \text{ adults.}$$

- 46 The sum of two numbers is 45, and their difference is 11. Find the two numbers.

Let the numbers be x and y with $x + y = 45$ and $x - y = 11$. Adding: $2x = 56 \Rightarrow x = 28$, then $y = 17$.

MCQ — Evaluating expressions

- 47 If $x = 6$, evaluate $4x - 9$.

a. 33

b. 24

c. -15

d. 15 ✓

$$4(6) - 9 = 24 - 9 = 15.$$

48 If $a = -2$ and $b = 3$, evaluate $a^2 + 2ab$.

a. -16

b. 8

c. -4

d. -8 ✓

$$a^2 = 4, 2ab = 2(-2)(3) = -12. 4 + (-12) = -8.$$

49 If $x = 5$ and $y = -1$, evaluate $(x + y)^2 - xy$.

a. 16

b. 11

c. 21 ✓

d. -5

$$(x + y)^2 = 4^2 = 16; xy = -5. 16 - (-5) = 21.$$

50 If $p = 3$, evaluate $2p^3 - 5p$.

a. 49

b. 39 ✓

c. 33

d. 69

$$2(27) - 15 = 54 - 15 = 39.$$

MCQ — Solving linear equations

51 Solve $\frac{x}{3} + 4 = 9$.

a. $x = 27$

b. $x = 15$ ✓

c. $x = 5$

d. $x = 39$

$$\frac{x}{3} = 5 \Rightarrow x = 15.$$

52 Solve $4(x + 2) = 3x + 15$.

a. $x = 1$

b. $x = 23$

c. $x = -7$

d. $x = 7$ ✓

$$4x + 8 = 3x + 15 \Rightarrow x = 7.$$

53 Solve $0.5x - 3 = 7$.

a. $x = 20$ ✓

b. $x = 8$

c. $x = -20$

d. $x = 14$

$$0.5x = 10 \Rightarrow x = 20.$$

54 Solve $\frac{2x - 1}{3} = 5$.

a. $x = 4$

b. $x = 7$

c. $x = 16$

d. $x = 8$ ✓

$$2x - 1 = 15 \Rightarrow 2x = 16 \Rightarrow x = 8.$$

55 Solve $6 - 2x = 3x - 9$.

a. $x = -3$

b. $x = 3$ ✓

c. $x = 15$

d. $x = 1$

$$15 = 5x \Rightarrow x = 3.$$

MCQ — Expanding and simplifying

56 Expand $(x + 7)^2$.

a. $x^2 + 7x + 49$

b. $x^2 + 14x + 14$

c. $x^2 + 14x + 49$ ✓

d. $x^2 + 49$

$$(x + 7)^2 = x^2 + 2(7)x + 49 = x^2 + 14x + 49.$$

57 Expand $(x - 5)^2$.

a. $x^2 + 10x + 25$

b. $x^2 - 5x + 25$

c. $x^2 - 10x + 25$ ✓

d. $x^2 - 25$

$$(x - 5)^2 = x^2 - 10x + 25.$$

58 Expand $(3x - 2)(3x + 2)$.

a. $9x^2 + 4$

b. $9x^2 - 12x - 4$

c. $9x^2 - 4$ ✓

d. $6x^2 - 4$

Difference of squares: $(3x)^2 - 2^2 = 9x^2 - 4$.

59 Simplify $2(x + 3) - 3(x - 1)$.

a. $5x + 9$

b. $-x + 9$ ✓

c. $x + 9$

d. $-x + 3$

$$2x + 6 - 3x + 3 = -x + 9.$$

MCQ — Factoring

60 Factor $3x^2 + 12x$.

a. $3(x^2 + 4x)$

b. $x(3x + 12)$

c. $3x(x + 12)$

d. $3x(x + 4)$ ✓

$$\text{GCF} = 3x: 3x^2 + 12x = 3x(x + 4).$$

61 Factor $x^2 - 11x + 30$.

a. $(x - 5)(x - 6)$ ✓

b. $(x - 2)(x - 15)$

c. $(x + 5)(x + 6)$

d. $(x - 3)(x - 10)$

Need two numbers multiplying to 30, adding to -11 : -5 , -6 .

62 Factor $x^2 + 3x - 18$.

a. $(x + 2)(x - 9)$

b. $(x - 6)(x + 3)$

c. $(x + 9)(x - 2)$

d. $(x + 6)(x - 3)$ ✓

Need two numbers multiplying to -18 , adding to 3: 6, -3 .

63 Factor $4x^2 - 25$.

a. $(2x - 25)(2x + 1)$

b. $(2x - 5)(2x + 5)$ ✓

c. $(2x - 5)^2$

d. $(4x - 5)(x + 5)$

Difference of squares: $(2x)^2 - 5^2 = (2x - 5)(2x + 5)$.

MCQ — Solving quadratic equations

64 Solve $x^2 - 9x + 18 = 0$.

- a. $x = -3$ or $x = -6$
- b. $x = 1$ or $x = 18$
- c. $x = 2$ or $x = 9$
- d. **$x = 3$ or $x = 6$ ✓**

$$(x - 3)(x - 6) = 0 \Rightarrow x = 3 \text{ or } x = 6.$$

65 Solve $x^2 + 2x - 24 = 0$.

- a. $x = 6$ or $x = -4$
- b. $x = -2$ or $x = 12$
- c. **$x = -6$ or $x = 4$ ✓**
- d. $x = -3$ or $x = 8$

$$(x + 6)(x - 4) = 0 \Rightarrow x = -6 \text{ or } x = 4.$$

66 The graph shows $y = x^2 - 49$. What are its roots (where the curve crosses the x-axis)?

- a. $x = 7$ only
- b. $x = 49$
- c. **$x = 7$ or $x = -7$ ✓**
- d. $x = -7$ only

The curve crosses the x-axis at $x = -7$ and $x = 7$, matching $x^2 - 49 = 0$.

67 Solve $2x^2 - 8x = 0$.

- a. $x = 8$
- b. $x = 2$ or $x = 4$
- c. **$x = 0$ or $x = 4$ ✓**
- d. $x = 0$ or $x = -4$

$$2x(x - 4) = 0 \Rightarrow x = 0 \text{ or } x = 4.$$

MCQ — Systems of two equations

68 Solve the system: $x + 2y = 8$, $x - y = 2$.

- a. $x = 2, y = 3$
- b. $x = 6, y = 1$
- c. $x = 4, y = 2$ ✓
- d. $x = 8, y = 0$

Subtracting: $3y = 6 \Rightarrow y = 2$, then $x = 4$.

69 Solve the system: $3x - y = 7$, $x + y = 5$.

- a. $x = 2, y = 3$
- b. $x = 1, y = 4$
- c. $x = 3, y = 2$ ✓
- d. $x = 4, y = 1$

Adding the equations: $4x = 12 \Rightarrow x = 3$, then $y = 2$.

70 The graph shows two lines. What is their point of intersection?

- a. (1, 4)
- b. (0, 5)
- c. (2, 3) ✓
- d. (3, 2)

The two lines $y = x + 1$ and $y = -x + 5$ cross where both are true: at (2, 3).

71 Solve the system: $y = 2x - 1$, $y = x + 3$.

a. $x = -4$, $y = -5$

b. $x = 2$, $y = 3$

c. $x = 7$, $y = 4$

d. $x = 4$, $y = 7$ ✓

$2x - 1 = x + 3 \Rightarrow x = 4$, then $y = 4 + 3 = 7$.

MCQ — Laws of exponents

72 Simplify $b^7 \div b^2$.

a. $b^{3.5}$

b. b^{14}

c. b^9

d. b^5 ✓

Subtract exponents: $b^{7-2} = b^5$.

73 Simplify $(2x^3)^2$.

a. $4x^6$ ✓

b. $2x^9$

c. $4x^5$

d. $2x^6$

$(2x^3)^2 = 2^2 x^{3 \times 2} = 4x^6$.

74 Simplify $x^0 \cdot x^4$ ($x \neq 0$).

- a. 1
- b. x^5
- c. x^0
- d. x^4 ✓

$x^0 = 1$, so $x^0 \cdot x^4 = x^4$.

75 Simplify m^{-2} .

- a. $-\frac{1}{m^2}$
- b. $\frac{1}{m^2}$ ✓
- c. m^2
- d. $-m^2$

A negative exponent means reciprocal: $m^{-2} = \frac{1}{m^2}$.

MCQ — Inequalities

76 The number line shows the solution set of an inequality. Which inequality matches?

- a. $x \geq 4$
- b. $x \leq 4$
- c. $x > 4$
- d. $x < 4$ ✓

An open circle at 4 with shading to the left means $x < 4$.

77 The number line shows the solution set of an inequality. Which inequality matches?

- a. $x \leq -2$
- b. $x < -2$
- c. $x \geq -2$ ✓
- d. $x > -2$

A closed circle at -2 with shading to the right means $x \geq -2$.

78 Solve $-3x + 7 \leq 22$.

- a. $x \geq -5$ ✓
- b. $x \leq -5$
- c. $x \leq 5$
- d. $x \geq 5$

$-3x \leq 15$; dividing by -3 flips the sign: $x \geq -5$.

79 The number line shows the solution set of an inequality. Which inequality matches?

- a. $x > 1$
- b. $x \geq 1$
- c. $x \leq 1$ ✓
- d. $x < 1$

A closed circle at 1 with shading to the left means $x \leq 1$.

MCQ — Reading function graphs

80 The graph shows a line. What is its slope?

a. $\frac{1}{2}$

b. 2 ✓

c. -2

d. -1

The line rises 2 for every 1 across, e.g. from (0, -1) to (1, 1): slope = 2.

81 What is the y-intercept of the line shown?

a. 4 ✓

b. 0

c. -4

d. 1

The line crosses the y-axis at (0, 4).

82 What is the equation of the line shown?

a. $y = 3x + 1$

b. $y = x + 3$

c. $y = \frac{1}{3}x$

d. $y = 3x$ ✓

The line passes through the origin and (1, 3): slope 3, no intercept, so $y = 3x$.

83 Find the slope of the line through the two marked points.

a. 2 ✓

b. $\frac{1}{2}$

c. -2

d. 4

$$\text{Slope} = \frac{6 - 2}{3 - 1} = \frac{4}{2} = 2.$$

MCQ — Substitution into formulas

84 Given $A = \frac{1}{2}bh$, find A when $b = 10$ and $h = 6$.

- a. 16
- b. 8
- c. 60
- d. 30 ✓

$$A = \frac{1}{2}(10)(6) = 30.$$

85 Given $KE = \frac{1}{2}mv^2$, find KE when $m = 4$ and $v = 3$.

- a. 18 ✓
- b. 12
- c. 24
- d. 36

$$KE = \frac{1}{2}(4)(9) = 18.$$

86 The rectangle shown has area $A = lw = 48$ and length $l = 8$. Find the width w .

- a. 8
- b. 6 ✓
- c. 384
- d. 40

$$48 = 8w \Rightarrow w = 6.$$

87 Given $F = ma$, find F when $m = 7$ and $a = 5$.

a. 35 ✓

b. 75

c. 12

d. 2.4

$$F = (7)(5) = 35.$$

MCQ — Translating words into equations

88 Three times a number, minus 5, equals twice the number plus 7. Find the number.

a. 19

b. 2

c. -12

d. 12 ✓

$$3n - 5 = 2n + 7 \Rightarrow n = 12.$$

89 The sum of two consecutive even integers is 54. Find the larger integer.

a. 28 ✓

b. 27

c. 29

d. 26

$$n + (n + 2) = 54 \Rightarrow n = 26; \text{ larger integer} = 28.$$

90 Five less than three times a number is 25. Find the number.

a. 10 ✓

b. 20

c. 5

d. 15

$$3n - 5 = 25 \Rightarrow n = 10.$$

91 The perimeter of a rectangle is 54 cm. The length is 3 cm more than the width. Find the width.

a. 24

b. 12 ✓

c. 15

d. 9

$$2(w + (w + 3)) = 54 \Rightarrow 4w + 6 = 54 \Rightarrow w = 12.$$

MCQ — Word problems

92 A number increased by 20% equals 72. Find the number.

a. 90

b. 52

c. 60 ✓

d. 86.4

$$1.2n = 72 \Rightarrow n = 60.$$

93 Ahmed is twice as old as his sister. In 5 years, the sum of their ages will be 40. Find Ahmed's current age.

a. 20 ✓

b. 10

c. 15

d. 25

Let sister = x , Ahmed = $2x$. $(x + 5) + (2x + 5) = 40 \Rightarrow 3x = 30 \Rightarrow x = 10$; Ahmed = 20.

94 A positive number plus its square is 30. Find the number.

a. 30

b. -6

c. 6

d. 5 ✓

$n^2 + n - 30 = 0 \Rightarrow (n + 6)(n - 5) = 0$; the positive root is $n = 5$.

95 The bar chart shows four students' test scores. Whose score is exactly 12 more than 70?

a. Student B

b. Student D

c. Student C ✓

d. Student A

$70 + 12 = 82$, which is Student C's score.

96 The product of two consecutive integers is 132. Find the larger integer.

- a. 13
- b. 12 ✓
- c. 10
- d. 11

$n(n+1) = 132 \Rightarrow n^2 + n - 132 = 0 \Rightarrow (n+12)(n-11) = 0$; taking the positive root $n = 11$, larger = 12.

Multiple choice — easy

97 Solve $9x - 15 = 3$.

- a. 2 ✓
- b. 4
- c. -2
- d. 0

$9x = 18$, so $x = 2$.

98 Solve $9x + 12 = -24$.

- a. -4 ✓
- b. -2
- c. 4
- d. -6

$9x = -36$, so $x = -4$.

99 Solve $8x + 13 = 117$.

a. -13

b. 13 ✓

c. 11

d. 15

$8x = 104$, so $x = 13$.

100 Solve $8x - 20 = -140$.

a. -17

b. -15 ✓

c. 15

d. -13

$8x = -120$, so $x = -15$.

101 Solve $9x - 12 = -147$.

a. -17

b. -15 ✓

c. 15

d. -13

$9x = -135$, so $x = -15$.

102 Solve $8x + 10 = 114$.

a. 13 ✓

b. 11

c. -13

d. 15

$8x = 104$, so $x = 13$.

103 Solve $9x + 15 = 150$.

a. 15 ✓

b. -15

c. 17

d. 13

$9x = 135$, so $x = 15$.

104 Solve $9x + 1 = 28$.

a. -3

b. 3 ✓

c. 5

d. 1

$9x = 27$, so $x = 3$.

105 Solve $2x = 6$.

a. 3 ✓

b. 5

c. 1

d. -3

$2x = 6$, so $x = 3$.

106 Solve $7x + 4 = -3$.

- a. 1
- b. 3
- c. -3
- d. -1 ✓

$$7x = -7, \text{ so } x = -1.$$

107 Solve $5x - 12 = -72$.

- a. -10
- b. -14
- c. -12 ✓
- d. 12

$$5x = -60, \text{ so } x = -12.$$

108 Solve $3x + 12 = 21$.

- a. -3
- b. 5
- c. 3 ✓
- d. 1

$$3x = 9, \text{ so } x = 3.$$

109 Solve $5x + 18 = -27$.

- a. -9 ✓
- b. 9
- c. -7
- d. -11

$$5x = -45, \text{ so } x = -9.$$

110 Solve $2x = -14$.

a. -7 ✓

b. -5

c. 7

d. -9

$2x = -14$, so $x = -7$.

111 Solve $5x - 10 = 20$.

a. 8

b. 4

c. 6 ✓

d. -6

$5x = 30$, so $x = 6$.

Multiple choice — medium

112 Expand $(x + 4)(x - 2)$.

a. $x^2 + 2x - 8$ ✓

b. $x^2 - 2x - 8$

c. $x^2 + 2x + 8$

d. $x^2 - 8x + 2$

$x^2 + (4 - 2)x + (4)(-2) = x^2 + 2x - 8$.

113 Expand $(x + 5)(x + 7)$.

- a. $x^2 - 12x + 35$
- b. $x^2 + 12x - 35$
- c. $x^2 + 12x + 35$ ✓
- d. $x^2 + 35x + 12$

$$x^2 + (5 + 7)x + (5)(7) = x^2 + 12x + 35.$$

114 Expand $(x + 3)(x + 7)$.

- a. $x^2 + 21x + 10$
- b. $x^2 - 10x + 21$
- c. $x^2 + 10x - 21$
- d. $x^2 + 10x + 21$ ✓

$$x^2 + (3 + 7)x + (3)(7) = x^2 + 10x + 21.$$

115 Expand $(x + 9)(x - 6)$.

- a. $x^2 + 3x - 54$ ✓
- b. $x^2 - 3x - 54$
- c. $x^2 - 54x + 3$
- d. $x^2 + 3x + 54$

$$x^2 + (9 - 6)x + (9)(-6) = x^2 + 3x - 54.$$

116 Expand $(x + 1)(x - 1)$.

- a. $x^2 + 2x - 1$
- b. $x^2 - x$
- c. $x^2 + 1$
- d. $x^2 - 1$ ✓

$$x^2 + (1 - 1)x + (1)(-1) = x^2 - 1.$$

117 Expand $(x + 2)(x + 2)$.

- a. $x^2 - 4x + 4$
- b. $x^2 + 6x + 4$
- c. $x^2 + 4x - 4$
- d. $x^2 + 4x + 4$ ✓

$$x^2 + (2 + 2)x + (2)(2) = x^2 + 4x + 4.$$

118 Expand $(x - 4)(x - 3)$.

- a. $x^2 + 12x - 7$
- b. $x^2 - 7x + 12$ ✓
- c. $x^2 + 7x + 12$
- d. $x^2 - 7x - 12$

$$x^2 + (-4 - 3)x + (-4)(-3) = x^2 - 7x + 12.$$

119 Expand $(x - 2)(x - 1)$.

- a. $x^2 + 2x - 3$
- b. $x^2 + 3x + 2$
- c. $x^2 - 3x + 2$ ✓
- d. $x^2 - 3x - 2$

$$x^2 + (-2 - 1)x + (-2)(-1) = x^2 - 3x + 2.$$

120 Expand $(x + 9)(x - 2)$.

- a. $x^2 - 7x - 18$
- b. $x^2 + 7x + 18$
- c. $x^2 - 18x + 7$
- d. $x^2 + 7x - 18$ ✓

$$x^2 + (9 - 2)x + (9)(-2) = x^2 + 7x - 18.$$

121 Expand $(x - 5)(x + 9)$.

- a. $x^2 + 4x - 45$ ✓
- b. $x^2 - 45x + 4$
- c. $x^2 + 4x + 45$
- d. $x^2 - 4x - 45$

$$x^2 + (-5 + 9)x + (-5)(9) = x^2 + 4x - 45.$$

122 Expand $(x - 2)(x - 8)$.

- a. $x^2 - 10x + 16$ ✓
- b. $x^2 + 10x + 16$
- c. $x^2 + 16x - 10$
- d. $x^2 - 10x - 16$

$$x^2 + (-2 - 8)x + (-2)(-8) = x^2 - 10x + 16.$$

123 Expand $(x - 4)(x + 7)$.

- a. $x^2 - 28x + 3$
- b. $x^2 + 3x + 28$
- c. $x^2 - 3x - 28$
- d. $x^2 + 3x - 28$ ✓

$$x^2 + (-4 + 7)x + (-4)(7) = x^2 + 3x - 28.$$

124 Expand $(x + 8)(x - 8)$.

a. $x^2 + 2x - 64$

b. $x^2 - 64$ ✓

c. $x^2 - 64x$

d. $x^2 + 64$

$$x^2 + (8 - 8)x + (8)(-8) = x^2 - 64.$$

125 Expand $(x + 4)(x - 3)$.

a. $x^2 - 12x + 1$

b. $x^2 + x + 12$

c. $x^2 - x - 12$

d. $x^2 + x - 12$ ✓

$$x^2 + (4 - 3)x + (4)(-3) = x^2 + x - 12.$$

126 Expand $(x - 1)(x - 1)$.

a. $x^2 - 2x + 1$ ✓

b. $x^2 - 2x - 1$

c. $x^2 + 2x + 1$

d. $x^2 + x - 2$

$$x^2 + (-1 - 1)x + (-1)(-1) = x^2 - 2x + 1.$$

Multiple choice — hard

127 Solve $x^2 - 9x + 14 = 0$.

- a. $-2, -7$
- b. $2, 8$
- c. $3, 7$
- d. $2, 7$ ✓

Factor: $(x - 2)(x - 7) = 0$, so $x = 2$ or $x = 7$.

128 Solve $x^2 - 14x + 48 = 0$.

- a. $6, 9$
- b. $-6, -8$
- c. $6, 8$ ✓
- d. $7, 8$

Factor: $(x - 6)(x - 8) = 0$, so $x = 6$ or $x = 8$.

129 Solve $x^2 - 25 = 0$.

- a. $5, -5$ ✓
- b. $-5, 5$
- c. $6, -5$
- d. $5, -4$

Factor: $(x - 5)(x + 5) = 0$, so $x = 5$ or $x = -5$.

130 Solve $x^2 - 12x + 35 = 0$.

- a. $8, 5$
- b. $-7, -5$
- c. $7, 5$ ✓
- d. $7, 6$

Factor: $(x - 7)(x - 5) = 0$, so $x = 7$ or $x = 5$.

131 Solve $x^2 + 10x + 16 = 0$.

- a. $-2, -7$
- b. $2, 8$
- c. $-2, -8$ ✓
- d. $-1, -8$

Factor: $(x + 2)(x + 8) = 0$, so $x = -2$ or $x = -8$.

132 Solve the system: $x - y = -1$, $x + y = -17$.

- a. $-8, -9$
- b. $-9, -8$ ✓
- c. $-9, -6$
- d. $-7, -8$

Adding the equations: $2x = -18$, so $x = -9$. Then $y = -8$.

133 Solve the system: $3x - y = -31$, $x + y = -5$.

- a. $4, -9$
- b. $-9, 6$
- c. $-7, 4$
- d. $-9, 4$ ✓

Adding the equations: $4x = -36$, so $x = -9$. Then $y = 4$.

134 Solve the system: $x + y = -3$, $x - y = 9$.

- a. $3, -4$
- b. $-6, 3$
- c. $3, -6$ ✓
- d. $5, -6$

Adding the equations: $2x = 6$, so $x = 3$. Then $y = -6$.

135 Solve the system: $x + y = -8$, $x - y = -10$.

- a. $1, -9$
- b. $-9, 1$ ✓
- c. $-9, 3$
- d. $-7, 1$

Adding the equations: $2x = -18$, so $x = -9$. Then $y = 1$.

136 Solve the system: $x - y = 5$, $x + y = -11$.

- a. $-3, -6$
- b. $-8, -3$
- c. $-1, -8$
- d. $-3, -8$ ✓

Adding the equations: $2x = -6$, so $x = -3$. Then $y = -8$.

137 Simplify $a^4 \times a^8$.

- a. a^{32}
- b. a^{11}
- c. a^{12} ✓
- d. a^{13}

$$a^{4+8} = a^{12}.$$

138 Simplify $\frac{x^9}{x^5}$.

a. x^5

b. x^4 ✓

c. x^3

d. x^{45}

$$x^{9-5} = x^4.$$

139 Simplify $(b^4)^3$.

a. b^{12} ✓

b. b^7

c. b^{14}

d. b^{10}

$$b^{4 \times 3} = b^{12}.$$

140 Simplify $a^4 \times a^5$.

a. a^8

b. a^{10}

c. a^{20}

d. a^9 ✓

$$a^{4+5} = a^9.$$

141 Simplify $\frac{x^7}{x^2}$.

a. x^{14}

b. x^5 ✓

c. x^6

d. x^4

$$x^{7-2} = x^5.$$